

Saketh Koona

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Education

Georgia Institute of Technology

BS in Industrial Engineering

May 2026

Atlanta, GA

- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Database Systems, Computer Vision

Technical Skills

Languages: Python, Elixir, SQL, Rust, TypeScript, C, C++, R, Java, Go

Technologies: Next.js, Git, Apache Spark, Apache Airflow, Kafka, OpenAI Gym, LangChain, AWS, Azure, GCP, Pandas, NumPy, Postgres

Experience

Software Engineer Intern

Jul 2026 - Present

Transpira Labs

Atlanta, GA

- Engineered core GTM platform with in-browser **WebRTC** softphone and **webhook** pipelines, automating outbound lead acquisition
- Architected an AI quoting engine parsing multi-channel streams (**Gmail, WhatsApp, Twilio, SMS**) via **LLM Tool Use**
- Built **DAT** and **Truckstop** ingestion and dispatch pipelines using **Next.js** and **Neon Postgres**
- Constructed **RLVR** environments and deterministic verifiers for reasoning benchmarks including **GDPVal, AccountingBench**

Open Source Contributor

Jun 2026 - Present

NeMo-RL

Atlanta, GA

- Contributed to **NVIDIA NeMo-RL** (open source), adding **800+** lines of unit and functional tests covering untested CLI entry points for checkpoint converters

Lead Data Chair

Apr 2025 - Dec 2025

Joint Finance Data Committee @ Georgia Tech

Atlanta, GA

- Leading a team of **15** in building and maintaining financial data systems to track **over 450** student organizations.
- Designed and deployed a **PostgreSQL** architecture on **AWS RDS** to manage over **\$25M** in annual transactions.
- Applied **XGBoost** and computer vision feature extraction to forecast and audit suspicious transactions, saving **\$13K**.
- Developed and maintained **ETL** workflows using **Apache Airflow** and **SQL** on **AWS EC2**, processing **12k** transactions monthly.

Lead Machine Learning Engineer

Aug 2024 - Present

AI-Based Innovation and Discovery Lab @ GT

Atlanta, GA

- Built a custom MDP and **reinforcement learning** environment for portfolio optimization, modeling realistic market conditions and defining states, masked actions, and action-to-weight mappings
- Implemented and trained **DDQN** based-network in **PyTorch** over **10,000+** episodes with risk-adjusted reward functions.
- Achieved **12.6%** average cumulative return and Sharpe ratio of **1.4** across backtests, demonstrating robust portfolio strategies.
- Fine-tuned LLMs on **100+** trading books using **LangChain** and **Ollama**, with **ROUGE-1: 0.68** and **BLEU: 0.53** benchmarks.

Machine Learning Student Researcher

Jan 2025 - May 2026

Structured Information for Precision Neuroengineering Lab (SIPLab)

Atlanta, GA

- Built custom **state-space architecture** in **PyTorch** to model latent brain dynamics across **256** neural channels.
- Trained **GRU-based recurrent networks** to simulate opsin behavior and capture nonlinear control dynamics, with **2%** reconstruction error.
- Generated seeded synthetic dataset pipeline with **Cleo** and **NumPy** for system identification, with **95%** eigenvalue recovery.
- Orchestrated large-scale, GPU-accelerated experiments via **Snakemake** and **SLURM** on HPC clusters with **H200 GPUs**.

Computer Vision Engineer

Aug 2024 - Jan 2025

Emory University Madabhushi Lab

Atlanta, GA

- Implemented **image segmentation** models and techniques to semantically distinguish noise from blood vessels in optical coherence tomography angiography (OCTA) retinal scans.
- Performed image processing/augmentation such as cropping, rescaling, and compression on over **150 OCTA-scans** using **OpenCV** and **NumPy**, improving model performance.
- Automated the generation of specific dataset structure to streamline model training using **Python**.
- Utilized multiple neural network architectures (**nn-Unet, OCTA-Net**) to detect relevant tubular structures in OCTA images, achieving a **97.6%** accuracy on initial training dataset.

Projects

Fake News Classification with LSTMs

- Built an **LSTM neural network** for fake news detection using **TensorFlow** and **Keras**.
- Reduced overfitting with **L2 regularization** and **dropout**, boosting test accuracy by **15%**.
- Cleaned and preprocessed text data with **Pandas** and applied **NLP (NLTK, spaCy)** for tokenization, lemmatization, and stopwords removal on **15K+ sequences**.
- Trained on **5K+ articles** from diverse sources, achieving **92.3% test accuracy**.
- Visualized key features using **Matplotlib**, including n-gram frequency and text length distributions.

Distributed Image Processing Engine | *Rust, Kafka, Axum, Docker, AWS S3, MongoDB, Tokio*

- Architected an event-driven **Rust** microservices pipeline using **Kafka** to asynchronously process large image workloads
- Decoupled ingress via an **Axum** API gateway with horizontally scaled, stateless worker pools using consumer groups
- Built a concurrent, topological **DAG scheduler** in **Tokio** to orchestrate multi-stage transformation tasks
- Implemented fault-tolerant persistence with **at-least-once processing** and idempotent writes to **AWS S3** and **MongoDB**

Volatility Forecasting and Trading | *Python, Pandas, Yfinance, PyTorch, TensorFlow, Matplotlib, NumPy*

- Built option chain data ingestion pipeline via **RestAPI** and used throttling for safe usage.
- Solved implied volatility via **Newton-Raphson** and **Black-Scholes Merton** in **NumPy** for **10,000** contracts.
- Predicted realized volatility for **30+** stocks with **LSTMs** in **TensorFlow** and **PyTorch**, achieving **15%** higher accuracy.
- Implemented a Python backtester tracking P&L, drawdown, and risk metrics over **2 years**.
- Delivered **4.57%** annualized excess returns with **Sharpe 1.30**.

Build: Scratch for Reinforcement Learning (1st Place, YC RL Hackathon) | *Python, HUD, React, FastAPI*

- Won **1st place** at the **HUD x YC RSI RL Hackathon**, earning a guaranteed **YC F26** interview.
- Built a **drag-and-drop** canvas compiling Environment/Tool/Task/Train blocks into runnable **RL environments**
- Built and shipped runnable templates spanning **tool use, reasoning, and classification** tasks, letting users build and train agents end-to-end.
- Turned the builder into its own **RL environment**, measuring agents' ability to compile block code into runnable **RL envs**.

C++ Market Exchange Simulator | *C++*

- Built a low-latency **price-time priority matching engine** and **internal orderbook**, supporting multiple order types.
- Implemented a lock-free **event-driven architecture** emitting deterministic order and trade events
- Built a **binary order entry protocol (BOE/OUCH-style)** with fixed-width messages, sequencing, and explicit acknowledgments
- Accepted order ingress via **TCP sessions** and published market data egress via **UDP multicast**
- Processed **12,000+** orders in single-threaded symbol partitions under synthetic load
- Optimized hot-path performance using **arena-based memory allocation** and object pooling

IMC Prosperity 3 Trading Bot | *Python, Pandas, Numpy*

- Built autonomous algo-trading bot for **IMC Prosperity 3**, placing in the **70th percentile** of **12,106** teams
- Built basket trading system for commodity assets, predicting relative price movements with **65%** accuracy
- Built order book monitoring tool using **Level 2 data** across **8+** price levels to detect order imbalances before price moves
- Implemented risk management capping position size per asset, limiting max drawdown to **12%**

Elixir AI-Benchmarker | *Elixir, Ecto, Typescript*

- Built distributed AI-evaluation benchmarking platform using **Oban** to execute **100+** evals concurrently
- Streamed real-time run progress to **Next.js** frontend over SSE, eliminating the need for polling
- Decoupled web and worker nodes through a shared **Postgres** job queue, enabling **horizontal scaling** by adding workers

What's Up Doc (Hackilytics 2025) | *Python, LangChain Flask, React.js, TypeScript*

- Built **RAG-powered chatbot** over **20,000+** medication records, mapping generic to brand-name drugs and surfacing **5-10** comparison options per query
- Utilized **FAISS vector database** for fast and efficient data indexing and retrieval.
- Built AI-powered patient caller dashboard; summarizing messages, triaging requests by urgency with email and call integration using **NLP**
- Designed web-app for user interaction with **React.js** frontend and **Flask** backend